ADMINISTRATIVE SERVICE DATA ANALYSIS

A personal data analytics case study using fictional administrative records

1,200	71.6%	4.7
Unique records	Completion rate	Avg. processing days

Prepared by
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Tools & methods
Microsoft Excel • Data Cleaning • Data Validation • Pivot-style Summaries • Reporting • Data Visualization

Project scope
Explore a fictional 2025 administrative-service dataset, address data-quality issues, summarize application activity, and present findings in a dashboard.

Transparency note: All records are fictional and created for portfolio practice. This is not official data and does not represent the performance of Dinas Sosial or any government agency.

01 — Business Problem & Dataset

Business question

How can administrative service records be organized and summarized to support routine monitoring of application volume, completion status, service demand, and processing time?

Objectives

• Assess and improve basic data quality.
• Summarize applications by month, district, service type, and status.
• Calculate completion rate and average processing time.
• Create a clear, reusable reporting view.

Dataset overview

Attribute	Description
Source	Synthetic / fictional practice data
Period	January–December 2025
Raw rows	1,208 (includes 8 duplicate records)
Unique records after cleaning	1,200
Fields	Record ID, applicant name, gender, age, district, service type, channel, date, status, processing

Data-quality issues intentionally included

Duplicate record IDs, missing district values, missing ages, inconsistent capitalization in service/status labels. These issues simulate common spreadsheet quality problems.

02 — Data Cleaning & Validation

The raw file was reviewed for duplicate identifiers, missing values, and inconsistent category labels. The cleaned sheet standardizes the fields used in the summary tables.

Cleaning actions

Issue	Treatment
Duplicate records	Removed duplicate rows using Record_ID; retained one row per ID.
Missing District	Replaced blank values with “Unknown” to preserve the record while making the gap visible.
Missing Age	Filled missing ages with the dataset median and converted age to an integer.
Inconsistent Service_Type	Standardized “social assistance” to “Social Assistance”.
Inconsistent Status	Standardized capitalization, e.g. “completed” → “Completed”.

Validation checks

• Confirmed unique Record_ID count after deduplication.
• Checked category values after standardization.
• Confirmed application dates fall within the 2025 reporting period.
• Recalculated summaries from the cleaned records.

Important limitation

Median imputation is used here as a simple practice technique. In a real project, the appropriate treatment for missing age values should depend on the business context and data-governance rules.

03 — Analysis & Key Findings

The following findings are calculated from the cleaned fictional dataset. They describe this dataset only.

1,200	**859**	**71.6%**
Total applications	Completed	Completion rate
4.7 days	**114**	**2025-03**
Average processing time	Pending applications	Highest-volume month

- **Application volume:** 2025-03 had the highest monthly volume (118 applications); 2025-06 had the lowest (73).
- **Service demand:** “Social Assistance” was the most common service category (378 records).
- **District volume:** “Semampir” had the highest application count (148 records) in this synthetic sample.
- **Processing time:** “Family Support” had the highest average processing time among service types (4.9 days).
- **Status mix:** 859 records were marked Completed; 158 In Progress; 114 Pending; and 69 Rejected.

Interpretation caution: Differences in counts or processing days are descriptive, not proof of service quality or cause. The dataset is randomly generated and is not representative of real applicants or districts.

04 — Recommendations & Next Steps

Recommendations for a real reporting workflow

1. **Monitor monthly volume:** use a consistent monthly report to identify workload changes and plan staffing or follow-up.

2. **Track status aging:** add days-in-current-status and a clear service-level target to distinguish normal processing from overdue cases.

3. **Improve data entry controls:** use dropdown lists, required fields, date validation, and unique record IDs to reduce inconsistent values.

4. **Define KPI rules:** document exactly what counts as “completed,” how completion rate is calculated, and whether the denominator includes rejected or withdrawn cases.

5. **Review missing data:** investigate why fields are missing before deciding whether to impute, exclude, or flag them.

Deliverables

• Excel workbook with Raw_Data, Cleaned_Data, summary sheets, and a Dashboard.
• This case-study PDF for portfolio presentation.

Skills demonstrated

Data cleaning • Data validation • Data quality • Excel reporting • KPI calculation • Descriptive analysis • Data visualization • Communicating limitations

Portfolio description (copy-ready)

Personal project using a fictional administrative-service dataset of 1,200 unique records. Cleaned duplicate and inconsistent records, handled missing values, analyzed monthly volume, service mix, status, and processing time, and built an Excel dashboard to communicate descriptive findings. All data is synthetic and does not represent any government agency.

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